

Домашняя работа 33 TMS по теме Google Cloud S3 Bucket + IAM.

1) Создание сервисного аккаунта с ролью Storage.Admin:

1.1) создаем сервис аккаунт

```
cloudshell:~ (it-server-344307)$ gcloud iam service-accounts create sa-tmsdz33 --display-name "TMS service account"  
Created service account [sa-tmsdz33].
```

1.2) добавляем сервис-аккаунту роль Storage.Admin

```
cloudshell:~ (it-server-344307)$ gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --member  
serviceAccount:sa-tmsdz33@$DEVSHHELL_PROJECT_ID.iam.gserviceaccount.com --role="roles/storage.admin"
```

```
Updated IAM policy for project [it-server-344307].  
bindings:  
- members:  
  - serviceAccount:service-52615052819@containerregistry.iam.gserviceaccount.com  
    role: roles/containerregistry.ServiceAgent  
- members:  
  - serviceAccount:52615052819-compute@developer.gserviceaccount.com  
  - serviceAccount:52615052819@cloudservices.gserviceaccount.com  
    role: roles/editor  
- members:  
  - user:egor.belousov@altezza.org  
    role: roles/owner  
- members:  
  - serviceAccount:gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com  
  - serviceAccount:sa-tmsdz33@it-server-344307.iam.gserviceaccount.com  
    role: roles/storage.admin  
version: 1
```

1.3) создаем ключ доступа

```
cloudshell:~ (it-server-344307)$ PROJECT_ID="$DEVSHHELL_PROJECT_ID" // ( it-server-344307 - ID проекта).  
cloudshell:~ (it-server-344307)$ SA_NAME="sa-tmsdz33"  
cloudshell:~ (it-server-344307)$ gcloud config set project it-server-344307 // указываем проект  
cloudshell:~ (it-server-344307)$ gcloud iam service-accounts keys create ~/keys/${SA_NAME}.json --iam-account=${SA_NAME}@${PROJECT_ID}.iam.gserviceaccount.com  
created key [eda440aa13d132d9538138b5d4fc28ab74d11678] of type [json] as [/home/egor_v/keys/sa-tmsdz33.json] for [sa-tmsdz33@it-server-344307.iam.gserviceaccount.com]
```

P.S.#1 или так gcloud iam service-accounts keys create key.json --iam-account=gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com (без переменных)

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts keys create key.json \  
--iam-account=gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com  
created key [1c1e5cad0bfe780800b579c73dc60971f0810611] of type [json] as [key.json] for [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
```

1.3.2) // создаем 2ю сервисную учетку "gitlab-runner-sa" (для теста) :

cloudshell:~ (it-server-344307)\$ **gcloud iam service-accounts create gitlab-runner-sa --description="Service account for GitLab runner on U22GITLABNODE2" --display-name="GitLab Runner SA"**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts create gitlab-runner-sa \
--description="Service account for GitLab runner on U22GITLABNODE2" \
--display-name="GitLab Runner SA"
Created service account [gitlab-runner-sa].
```

1.3.3) Проверяем созданные IAM-учетки в нашем проекте :

cloudshell:~ (it-server-344307)\$ **gcloud iam service-accounts list**

DISPLAY NAME: Compute Engine default service account
EMAIL: 52615052819-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME: GitLab Runner SA
EMAIL: gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
DISABLED: False

DISPLAY NAME: TMS service account
EMAIL: sa-tmsdz33@it-server-344307.iam.gserviceaccount.com
DISABLED: False

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud iam service-accounts list
DISPLAY NAME: Compute Engine default service account
EMAIL: 52615052819-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME: GitLab Runner SA
EMAIL: gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com
DISABLED: False

DISPLAY NAME: TMS service account
EMAIL: sa-tmsdz33@it-server-344307.iam.gserviceaccount.com
DISABLED: False
egor_belousov@cloudshell:~ (it-server-344307)$
```

1.4) копируем ключ (json-файл) на удаленный хост:

@cloudshell:~/keys (it-server-344307)\$ **cat sa-tmsdz33.json**

```
{
  "type": "service_account",
```

```

"project_id": "it-server-344307",
"private_key_id": "eaa13d132d9538138b5d4fc28ab74d11678",
"private_key": "-----BEGIN PRIVATE
KEY-----\nMIIEvwlBADANBgkqhkiG9w0BAQEFAASCBAkwggSIAgEAAoIBAQDAPRIJ+/yV79B1\nnbZOTJz5oXHfpayfKT06PBOdOibM97OTFKvCmA94dXRDlL2MIQWuOW4RKe8wK6ZL\nncmUr97/f9spqQ/pksN4Kw
AUXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXQcuNGbg==\n-----END PRIVATE KEY-----\n",
"client_email": "sa-tmsdz33@it-server-344307.iam.gserviceaccount.com",
"client_id": "115169803436673907629",
"auth_uri": "https://accounts.google.com/o/oauth2/auth",
"token_uri": "https://oauth2.googleapis.com/token",
"auth_provider_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs",
"client_x509_cert_url": "https://www.googleapis.com/robot/v1/metadata/x509/sa-tmsdz33%40it-server-344307.iam.gserviceaccount.com",
"universe_domain": "googleapis.com"
}

```

1.5) авторизоваться на ubuntu22 по key.json-файлу :

```
@cloudshell:~/keys (it-server-344307) gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
```

Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]

```
root@U22GITLABNODE2:/home/gitlab-runner# nano key.json
root@U22GITLABNODE2:/home/gitlab-runner# cat key.json
{
  "type": "service_account",
  "project_id": "it-server-344307",
  "private_key_id": "1c1e5cad0bfe780800b579c73dc60971f0810611",
  "private_key": "-----BEGIN PRIVATE KEY-----\nMIIeVAIBADANBgkqhkiG9w0BAQEFAASCBKYYggSiAgEAAoIBAQD3Us4L\nb2pa9YRAXVl3vgdyd4DM/nwHCTrhS1uTuC9bdyYsiVy9V982CrqpsrO6glCzwhlFIgQ4dwLTgHPe/6iAH5X9\ndJbPLYByPK3PqK0:\njyEPBAGMBAAECggAEEDUBStxeSjn3xJA34CG7sPSvQL5e+bfMODZ96VKohtzn6+fvm8/tlCRu+n/u9aoWovfvny4wrKRggbrZr4S2I\nFh6pUvVm+7qx2QHq/WfoKZgEt7xc45G7z5Az\ntp057TxxhoP19ohmdh+s/kkyUAzbyf7ht+CrzyvzqjALYd8nwmp5Gs8o4FJKpc7f\\  
6TcdCVo4tRQ/nbwNu3ixop/rRk0aoF7bbfHPKYm+ur4nikzeYd80iyfIpi69k5llSa2+sTEdsbbvnKmCdr/24busSIjvkjnfV++ioAQKBGU/LT/np1IUBGSVLBQTAY/PpSnRom8Kxj+w\\niS1LRKEIECa4em9yb/z/fjcJQK0fkLaU/yef2bTuG09usb/idvzg/CylcyxJHYB\\nSGakZEBI67sL\nvaNDEq6JMguHDfoq9D0/nwUAFkl1qzakis/lyIKhq+7RPzwOpYZ\\nen2o67cSGQKBgdgtitCXChq4sJton1f3+SRY+/c7ql7vjRXqsDe+WYrhN94Mjc5MZ\\nb+8AG3gQE0/8NNP5rchhlOEfpbnza7qBY06wzVMiYOYAWsjsBMupNqax05khzhln8vgKCy/UghM8THTCWbnis9reJu5/D1FS08aQ\nAOWxLCf5K3qlg6cu/gXAogAA8qi\\npws/kvyH7QA864tbj09Xj9/XMJ16RS21x2AX14VJBmo4KxmhcL6jiw0+cD6epdbunl7oljftw3Pak9ojZYga8xVqqBfiuwJwnaz29hkvnZo4oe+CKDcy9eIEVE7iT4GLE\\nxQow5yasw7ra317v+kArKXe/HA8vgxBR5awsWBECgyA3ed8vMaouwUbck5LCjYH\nMZN\\neiha0waz8ADIewnOGTWycmfITouXZKPpxG6noS4U4FL0MD5KpXZnuSUQn6cwjuPTWI7v5S2ZcNEBeIO/RJTJniaffL2xs0FNZdvfb9uq/McfGde\\nprUlF2IO37ltLZDA0mqw+w==\\n-----END PRIVATE KEY-----\\n",
  "client_email": "gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com",
  "client_id": "114450695213956486347",
  "auth_uri": "https://accounts.google.com/o/oauth2/auth",
  "token_uri": "https://oauth2.googleapis.com/token",
  "auth_provider_x509_cert_url": "https://www.googleapis.com/oauth2/v1/certs",
  "client_x509_cert_url": "https://www.googleapis.com/robot/v1/metadata/x509/gitlab-runner-sa%40it-server-344307.iam.gserviceaccount.com",
  "universe_domain": "googleapis.com"
}
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account egor.belousov@altezza.org --key-file=key.json
ERROR: (gcloud.auth.activate-service-account) Invalid value for [ACCOUNT]: The given account name does not match the account name in the key file. This argument can be omitted when using .json keys.
root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
root@U22GITLABNODE2:/home/gitlab-runner#
```

1.6) проверяем наш сервис-аккаунт в Google Cloud :

IAM & Admin

/ Service accounts

/ Service account: 115169803436673907629

IAM

PAM

Principal Access Boundary

Security Insights

Identity & Organization

Policy Troubleshooter

Policy Analyzer

Organization Policies

Service Accounts

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TMS service account

Details

Permissions

Keys

Metrics

Logs

Principals with access

Service account details

Name

TMS service account

Save

Description

Save

Email

sa-tmsdz33@it-server-344307.iam.gserviceaccount.com

Unique ID

115169803436673907629

Service account status

Disabling your account allows you to preserve your policies without having to delete it.

Enabled

Disable service account

Advanced settings

Domain-wide Delegation

Granting this service account access to your organization's data via domain-wide delegation should be used with caution. It can be reversed by disabling or deleting the service account or by removing access through the Google Workspace admin console.

Learn more about domain-wide delegation

IAM & Admin

/ Service accounts

/ Service account: 115169803436673907629

/ Keys

IAM

PAM

Principal Access Boundary

Security Insights

Identity & Organization

Policy Troubleshooter

Policy Analyzer

Organization Policies

Service Accounts

Workload Identity Federation

Workforce Identity Federati...

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Privacy & Security

Identity-Aware Proxy

TMS service account

Details

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Keys

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Logs

Principals with access

Keys

Service account keys could pose a security risk if compromised. We recommend you avoid downloading service account keys from public repositories. [Learn more about Google Cloud](#)

Google automatically disables service account keys detected in public repositories. You can customize this behavior by using [organization policies](#)

Add a new key pair or upload a public key certificate from an existing key pair.

Block service account key creation using [organization policies](#)

[Learn more about setting organization policies for service accounts](#)

Add key

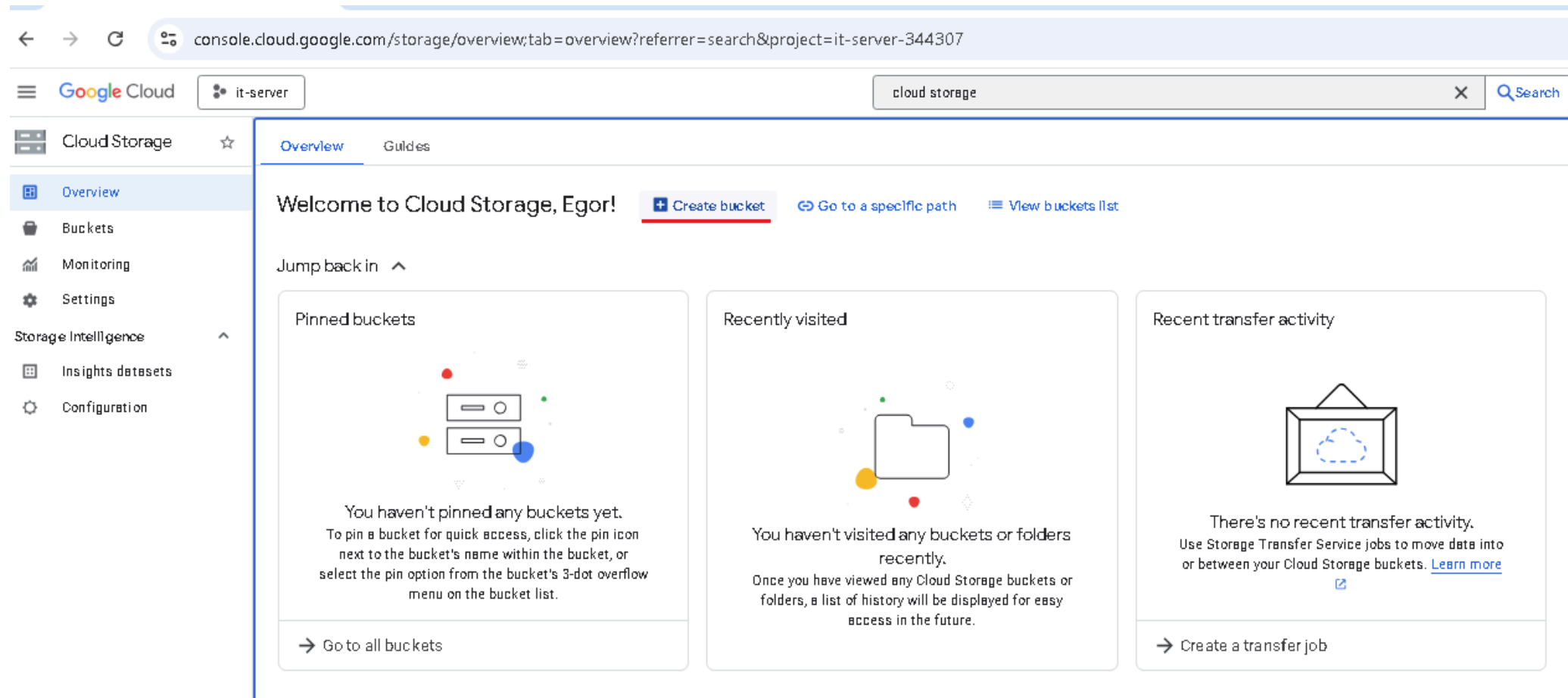
Type	Status	Key	Creation date	Expiration date	
	Active	eda440aa13d132d9538138b5d4fc28ab74d11678	Feb 12, 2026	Jan 1, 10000	

2) Создаем S3 GCP Bucket (cli-вариант)

cloudshell:~ (it-server-344307)\$ **gcloud storage buckets create gs://tmsdz33 --location=EUROPE-WEST2 --default-storage-class=STANDARD**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets create gs://tmsdz33 --location=EUROPE-WEST2 --default-storage-class=STANDARD
Creating gs://tmsdz33/...
```

2.1) Создаем bucket (gui-вариант создания) :



←

→

↺

console.cloud.google.com/storage/create-bucket?project=it-server-344307

Google Cloud

it-server

cloud storage

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Configuration

←

Create a bucket

•

Get Started

Pick a globally unique, permanent name. [Naming guidelines](#)

tmsdz33

Tip: Don't include any sensitive information

Labels (optional)

Labels are key:value pairs that allow you to group related buckets together or with other Cloud Platform resources. [Learn more](#)

Key 1

Value 1

descriptonbucket

tmsdz33

+ Add label

Continue

•

Choose where to store your data

Location: us (multiple regions in United States)

Location type: Multi-region

Good to know

Location pricing

Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Multi-region / Standard

Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

[Estimate your monthly cost](#)

←

→

↺

console.cloud.google.com/storage/create-bucket?project=it-server-344307

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Create a bucket

✓

Get Started

Name: tmsdz33

✓

Choose where to store your data

This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. [Learn more](#)

Location type

☐ Multi-region

Highest availability across largest area

☐ Dual-region

High availability and low latency across 2 regions

☒ Region

Lowest latency within a single region

europa-central2 (Warsaw)

☐ Add cross-bucket replication via Storage Transfer Service

As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. [Learn more](#)

Continue

Good to know

Location pricing

Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Region / Standard

Item	Cost
europa-central2 (Warsaw)	\$0.023 per GB-month

[Estimate your monthly cost](#)

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cloud storage

X

Cloud Storage

☆

← Create a bucket

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✓ Choose where to store your data

Location: europe-central2 (Warsaw)

Location type: Region

• Choose how to store your data

A storage class sets costs for storage, retrieval, and operations, with minimal differences in uptime. Choose if you want objects to be managed automatically or specify a default storage class based on how long you plan to store your data and your workload or use case. [Learn more](#)

☐ Autoclass

Automatically transitions each object to Standard or Nearline class based on object-level activity to optimize for cost and latency. Recommended if usage frequency may be unpredictable. Can be changed to a default class at any time. [Pricing details](#)

☒ Set a default class

Applies to all objects in your bucket unless you manually modify the class per object or set object lifecycle rules. Best when your usage is highly predictable.

☒ Standard

Best for short-term storage and frequently accessed data

☐ Nearline

Best for backups and data accessed less than once a month

☐ Coldline

Best for disaster recovery and data accessed less than once a quarter

☐ Archive

Best for long-term digital preservation of data accessed less than once a year

Optimize storage for data-intensive workloads

☐ Enable Hierarchical namespace on this bucket

Optimize for AI/ML and analytics with a filesystem-like hierarchical structure. This permanent option enables atomic folder renames, faster folder listing, and other enhancements not available in Cloud Storage's standard buckets that uses flat namespace.

location of your bucket: [Pricing details](#)

Current configuration: Region / Standard

Item	Cost
europe-central2 (Warsaw)	\$0.023 per GB-month

[Estimate your monthly cost](#)

Cloud Storage ☆

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- Storage Intelligence ^
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 - Configuration

- Marketplace
- Release Notes
- <|

← Create a bucket

✓ Get Started

Name: tmsdz33

✓ Choose where to store your data

Location: europe-central2 (Warsaw)

Location type: Region

✓ Choose how to store your data

Default storage class: Standard

Hierarchical namespace: Disabled

Anywhere Cache: Disabled

• Choose how to control access to objects

Prevent public access

Restrict data from being publicly accessible via the internet. Will prevent this bucket from being used for web hosting. [Learn more](#)

☒ Enforce public access prevention on this bucket

Access control

☒ **Uniform**
Ensure uniform access to all objects in the bucket by using only bucket-level permissions (IAM). This option becomes permanent after 90 days. [Learn more](#)

☐ Fine-grained
Specify access to individual objects by using object-level permissions (ACLs) in addition to your bucket-level permissions (IAM). [Learn more](#)

Continue

• Choose how to protect object data

Soft delete policy: Default

Object versioning: Disabled

Bucket retention policy: Disabled

Object retention: Disabled

Encryption type: Google-managed

Create

Cancel

Good to know

Location pricing

Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Region / Standard

Item	Cost
europe-central2 (Warsaw)	\$0.023 per GB-month

[Estimate your monthly cost](#)

console.cloud.google.com/storage/create-bucket?project=it-server-344307

Google Cloud

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Create a bucket

✓

Choose where to store your data

Location: europe-central2 (Warsaw)

Location type: Region

✓

Choose how to store your data

Default storage class: Standard

Hierarchical namespace: Disabled

Anywhere Cache: Disabled

✓

Choose how to control access to objects

Public access prevention: On

Access control: Uniform

•

Choose how to protect object data

Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to add extra layers of security.

Data protection

☒

Soft delete policy (For data recovery)

When enabled, this bucket and its objects will be kept for a specified period after they're deleted and can be restored during this time. [Learn more](#)

☒

Use default retention duration

All buckets have a 7 day soft delete duration by default, unless this default has been customized by your organization administrator.

☐

Set custom retention duration

Specify how long this bucket and its objects should be retained after they're deleted. Setting a '0' duration disables soft delete, meaning any deleted objects will be permanently deleted.

☐

Object versioning (For version control)

For restoring deleted or overwritten objects. To minimize the cost of storing versions, we recommend limiting the number of noncurrent versions per object and scheduling them to expire after a number of days. [Learn more](#)

☐

Retention (For compliance)

For preventing the deletion or modification of the bucket's objects for a specified period of time.

▼

Data Encryption

Create

Cancel

location of your bucket. [Pricing details](#)

Current configuration: Region / Standard

Item	Cost
europe-central2 (Warsaw)	\$0.023 per GB-month

[Estimate your monthly cost](#)

2.2) Предоставляем доступ к бакету по email :

```
cloudshell:~ (it-server-344307)$ gcloud storage buckets add-iam-policy-binding gs://tmsdz33 --member="user:egor.belousov@altezza.org" --role="roles/storage.objectAdmin"
```

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets add-iam-policy-binding gs://tmsdz33 --member="user:egor.belousov@altezza.org" --role="roles/storage.objectAdmin"
bindings:
- members:
  - projectEditor:it-server-344307
  - projectOwner:it-server-344307
  role: roles/storage.legacyBucketOwner
- members:
  - projectViewer:it-server-344307
  role: roles/storage.legacyBucketReader
- members:
  - user:egor.belousov@altezza.org
  role: roles/storage.objectAdmin
etag: CAU=
kind: storage#policy
resourceId: projects/_/buckets/tmsdz33
version: 1
egor_belousov@cloudshell:~ (it-server-344307)$
```

2.3) Проверяем атрибуты нашего бакета "tmsdz33" :

```
cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33
```

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33
acl:
- entity: project-viewers-52615052819
  projectTeam:
    projectNumber: '52615052819'
    team: viewers
  role: READER
default_storage_class: STANDARD
generation: 1770038782129218882
location: EUROPE-WEST2
location_type: region
metageneration: 5
name: tmsdz33
public_access_prevention: inherited
satisfies_pzs: true
soft_delete_policy:
  effectiveTime: '2026-02-02T13:26:22.487000+00:00'
  retentionDurationSeconds: '604800'
storage_url: gs://tmsdz33/
uniform_bucket_level_access: false
update_time: 2026-02-02T17:53:24+0000
egor_belousov@cloudshell:~ (it-server-344307)$
```

// можно смотреть атрибуты бакета " tmsdz33 " в json-формате:

cloudshell:~ (it-server-344307)\$ **gcloud storage buckets describe gs://tmsdz33 --format="json(name,location,storageClass)"**

```
egor_belousov@cloudshell:~ (it-server-344307)$ gcloud storage buckets describe gs://tmsdz33 --format="json(name,location,storageClass) "
{
  "location": "EUROPE-WEST2",
  "name": "tmsdz33"
}
egor_belousov@cloudshell:~ (it-server-344307)$
```

2.4) Проверяем свойства бакета через GUI

Google Cloud

it-server

cloud storage

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tmsdz33

Location

Storage class

Public access

Protection

Location

europa-central2 (Warsaw)

Storage class

Standard

Public access

Not public

Protection

Soft Delete

Objects

Configuration

Permissions

Protection

Lifecycle

Observability

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Overview

Created

February 2, 2026, 1:35:43 PM GMT+3

Updated

February 2, 2026, 1:35:43 PM GMT+3

Hierarchical namespace

Not enabled

Location type

Region

Location

europa-central2 (Warsaw)

Replication

—

Cross-bucket replication

Not enabled

Default storage class

Standard

Requester Pays

Off

Tags

None

Labels

descriptionbucket:tmsdz33

Cloud Console URL

https://console.cloud.google.com/storage/browser/tmsdz33

gsutil URI

gs://tmsdz33

Cross-origin resource sharing

Not enabled

Permissions

Access control

Uniform

Public access prevention

Enabled via bucket setting

Public access status

Not public

IP filtering

Not configured

Protection

Soft delete policy

7 days

Object versioning

Off

Bucket retention policy

None

Object retention

Disabled

Encryption type

Google-managed

Object lifecycle

Lifecycle rules

None

Anywhere Cache

Cache

Not supported

Compute zones

—

3) Подключаемся к google cloud s3 bucket с ubuntu :

```
ubuntu22# snap install google-cloud-cli --classic
```

```
// пакет для работы с Google Cloud API на ubuntu
```

```
ubuntu22# gcloud auth login
```

// выдет https-строку-запрос-на-токен идем в браузер и авторизация через свою email-учетку.

Please check that your browser is logged in as account [redacted] and that you are using the correct browser profile.
white:~# gcloud auth login
Go to the following link in your browser, and complete the sign-in prompts:

https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fwww.googleapis.com%2Faauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Faauth%2Fclogin-platform-engine.admin+https%3A%2F%2Fwww.googleapis.com%2Faauth%2Fsamlserver.compute+https%3A%2F%2Fwww.googleapis.com%2Faauth%2Fsamlserver.d=S256

Once finished, enter the verification code provided in your browser: 4/0Asc3gC3DsbnMgPhw7ooqekMJ3st4gmQAfro8wdGGe523Lrx5tfDH-IWqnI

You are now logged in as [redacted].
Your current project is [it-server-344307]. You can change this setting by running:

```
$ gcloud config set project PROJECT_ID
```

P.S.#1 Можно и без браузера через **"gcloud auth login --no-browser"** но до конца не разобрался как.

```
ubuntu22# gcloud config set project it-server-344307
```

// где it-server-344307 - ID проекта).

```
white:~# gcloud config set project it-server-344307
Project 'it-server-344307' lacks an 'environment' tag. Please create or add a tag with key 'environment' and a value like 'production', 'development', 'test', or 'staging'. Add an 'environment' tag using 'gcloud resource-manager tags bindings create'. See https://cloud.google.com/resource-manager/docs/creating-managing-projects#designate\_project\_environments\_with\_tags for details.
updated_property [core/project].
```

```
ubuntu22#gcloud storage buckets create gs://tmsdz331 --location="europe-west2"
```

Creating gs://tmsdz331/...

3.1) Проверяем какие бакеты создались через cli :

Buckets

 Create

 Refresh

Filter Filter buckets

②

☐	Name ↑	Created	Location type	Location	Default storage class ?	Last modified	Public access ?
☐	tmsdz33	Feb 2, 2026, 4:26:22 PM	Region	europe-west2	Standard	Feb 3, 2026, 12:13:15 AM	Subject to object ACLs
☐	tmsdz331	Feb 8, 2026, 11:40:22 PM	Region	europe-west2	Standard	Feb 8, 2026, 11:40:22 PM	Subject to object ACLs

3.2) Копируем все файлы из папки /home/gitlab-runner/testsite/ в бакет “ tmsdz33 “ :

Ubuntu22# `gcloud storage cp * gs://tmsdz33`

```
root@U22GITLABNODE2:/home/gitlab-runner/testsite#
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 1.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 2.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# nano 3.txt
root@U22GITLABNODE2:/home/gitlab-runner/testsite# gcloud storage cp * gs://tmsdz33
Copying file:///1.txt to gs://tmsdz33/1.txt
Copying file:///2.txt to gs://tmsdz33/2.txt
Copying file:///3.txt to gs://tmsdz33/3.txt
Copying file:///index.html to gs://tmsdz33/index.html
Completed files 4/4 | 60.0B/60.0B

Average throughput: 247.6B/s
root@U22GITLABNODE2:/home/gitlab-runner/testsite#
```

3.3) Проверяем в GUI GCP что файлы успешно скопировались в бакет:

it-server

buckets

Search

2

☆

← Bucket details

Edit bucketGo to pathRefresh

tmsdz33

Location: europe-west2 (London)Storage class: StandardPublic access: Subject to object ACLsProtection: Soft Delete

Objects

Configuration

Permissions

Protection

Lifecycle

Observability

Inventory Reports

Operations

Folder browser

Buckets > tmsdz33

Create folderUploadTransfer dataOther servicesLearn

Filter by name prefix onlyFilterFilter objects and foldersShowLive objects only

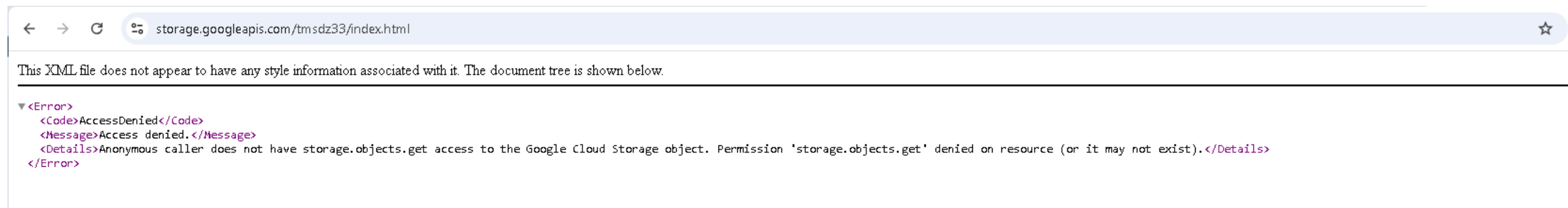
☐	Name	Size	Type	Created	Storage class	Last modified	Public access	Version history	
☐	1.txt	4 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public	—	DownloadMore
☐	2.txt	5 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public	—	DownloadMore
☐	3.txt	9 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public	—	DownloadMore
☐	index.html	42 B	text/html	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public	—	DownloadMore

3.4) Копируем файлы во 2й бакет "tmsdz331" :

```
cloudshell:~ (it-server-344307)$ gcloud storage cp * gs://tmsdz331
root@U22GITLABNODE2:/home/gitlab-runner/testsite# gcloud storage cp * gs://tmsdz331
Copying file:///1.txt to gs://tmsdz331/1.txt
Copying file:///2.txt to gs://tmsdz331/2.txt
Copying file:///3.txt to gs://tmsdz331/3.txt
Copying file:///index.html to gs://tmsdz331/index.html
Completed files 4/4 | 60.0B/60.0B

Average throughput: 214.6B/s
```

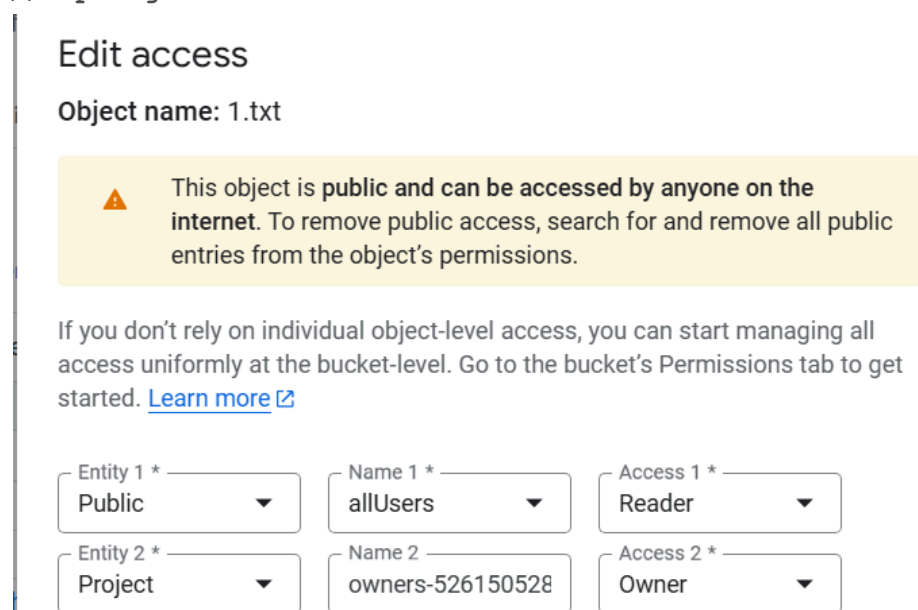
3.5) Проверяем доступ из Интернета, его нет , так как при создании указали бакет без доступа из Интернета (без public доступа) поэтому все норм, доступ только через key.json.



3.6) Открываем доступ из Интернета к конкретному файлу в S3 bucket :

```
cloudshell:~ (it-server-344307)$ gsutil acl ch -u AllUsers:R gs://tmsdz33/1.txt
```

//Через gui GCP:



Objects

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tmsdz33

Buckets > tmsdz33

Create folder

Upload

Transfer data

Other services

Learn

Filter by name prefix only

Filter

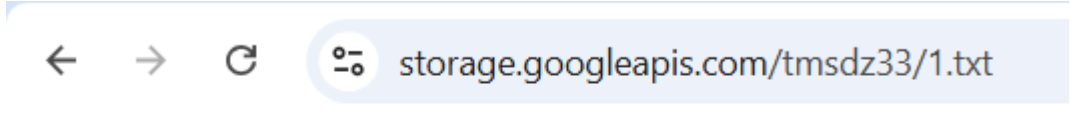
Filter objects and folders

Show

Live objects only

☐	Name	Size	Type	Created	Storage class	Last modified	Public access		
☐	 1.txt	4 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 9:01:51 PM	 Access granted to public principals	Copy URL	 
☐	 2.txt	5 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public		 
☐	 3.txt	9 B	text/plain	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public		 
☐	 index.html	42 B	text/html	Feb 2, 2026, 8:40:39 PM	Standard	Feb 2, 2026, 8:40:39 PM	Not public		 

3.7) Проверяем в браузере доступ к файлу 1.txt (открыт на чтение для всего Интернета) :



123

3.8) Удаляем из бакета s3://tmsdz33 файл 3.txt

```

root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]

root@U22GITLABNODE2:/home/gitlab-runner# gcloud storage rm gs://tmsdz33/3.txt
Removing objects:
Removing gs://tmsdz33/3.txt...
Completed 1/1

root@U22GITLABNODE2:/home/gitlab-runner# gcloud auth activate-service-account gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com --key-file=key.json
Activated service account credentials for: [gitlab-runner-sa@it-server-344307.iam.gserviceaccount.com]
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner#
root@U22GITLABNODE2:/home/gitlab-runner# gcloud storage rm gs://tmsdz33/3.txt
Removing objects:
Removing gs://tmsdz33/3.txt...
Completed 1/1
root@U22GITLABNODE2:/home/gitlab-runner#

```